PHYS 320 Examination

Thermodynamics and Statistical Mechanics – Answer Key

1. (C) Gibbs free energy G.
At constant temperature and pressure (most common lab conditions), the Gibbs free energy $G = H - TS$ is minimized at equilibrium. The condition $dG = 0$ determines spontaneity: $\Delta G < 0$ for spontaneous processes.

2. (B) $TV^{\gamma-1} =$ constant.
For a reversible adiabatic process ($Q = 0$), the relationship $TV^{\gamma-1} =$ constant holds, where $\gamma = C_P/C_V$ is the heat capacity ratio. Equivalently, $PV^{\gamma} =$ constant.

3. (C) A sum over Boltzmann factors for all states, encoding thermodynamic information.
The partition function $Z = \sum_i e^{-\beta E_i}$ (where $\beta = 1/k_B T$) is the central object in statistical mechanics. All thermodynamic quantities can be derived from Z, including free energy $F = -k_B T \ln Z$.

4. (A) $\eta = 1 - \frac{T_C}{T_H}$.
The Carnot efficiency represents the maximum possible efficiency for any heat engine operating between two reservoirs. It depends only on the temperatures and sets a fundamental limit based on the second law of thermodynamics.

5. *Sample answer:*
Zeroth Law: If two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other. This establishes the concept of temperature as a transitive property.

First Law: $\Delta U = Q - W$ (energy conservation). The change in internal energy equals heat absorbed minus work done. This states that energy cannot be created or destroyed, only converted.

Second Law: In an isolated system, entropy never decreases: $\Delta S \geq 0$. This gives directionality to time and explains why certain processes are irreversible. It limits the efficiency of heat engines and prohibits perpetual motion machines of the second kind.

Third Law: As temperature approaches absolute zero, the entropy of a perfect crystal approaches zero: $S \to 0$ as $T \to 0$. This provides an absolute scale for entropy and implies that absolute zero is unattainable.

Together, these laws constrain energy conversion, establish equilibrium conditions, and explain irreversibility in nature.

6. *Sample answer:*
Boltzmann's formula: $S = k_B \ln \Omega$, where Ω is the number of accessible microstates corresponding to a macrostate.

Derivation concept:

- A macrostate with more microstates is more probable
- Entropy must be additive for independent systems: if system 1 has Ω_1 states and system 2 has Ω_2 states, the combined system has $\Omega = \Omega_1 \times \Omega_2$ states

- Additivity requires $S(\Omega_1 \times \Omega_2) = S(\Omega_1) + S(\Omega_2)$, which is satisfied by the logarithm: $\ln(\Omega_1 \times \Omega_2) = \ln \Omega_1 + \ln \Omega_2$

Directionality: Spontaneous processes evolve toward macrostates with larger Ω (more microstates), increasing entropy. This explains why systems naturally progress from ordered to disordered states and why time has a preferred direction (thermodynamic arrow of time).

7. *Sample answer:*
Isothermal process (constant T):

- For ideal gas: $\Delta U = 0$ (internal energy depends only on T)
- $Q = W = nRT \ln(V_f/V_i)$ for expansion
- Entropy increases: $\Delta S = nR \ln(V_f/V_i) > 0$ for expansion

Adiabatic process ($Q = 0$):

- Temperature decreases during expansion: $T_f < T_i$
- $\Delta U = -W$ (work comes from internal energy)
- Entropy remains constant: $\Delta S = 0$ (reversible adiabatic process)
- The gas cools because it does work against external pressure without heat input

Key difference: Isothermal processes maintain temperature via heat exchange; adiabatic processes are thermally isolated, so temperature changes as work is done.

8. *Sample answer:*
Chemical potential: μ is the change in Gibbs free energy when one particle is added to the system at constant T and P: $\mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}$.

Phase equilibrium condition: At equilibrium between phases (e.g., liquid and vapor), the chemical potentials must be equal: $\mu_{\text{liquid}} = \mu_{\text{vapor}}$.

Physical meaning:

- If $\mu_{\text{liquid}} > \mu_{\text{vapor}}$, particles spontaneously move from liquid to vapor (evaporation) to lower the total free energy
- If $\mu_{\text{liquid}} < \mu_{\text{vapor}}$, condensation occurs
- Equilibrium is reached when μ is uniform across all phases

Example - Vapor-liquid coexistence: At a given temperature, the vapor pressure adjusts until $\mu_{\text{liquid}}(T, P) = \mu_{\text{vapor}}(T, P)$. This determines the saturation vapor pressure and the coexistence curve in the phase diagram.